

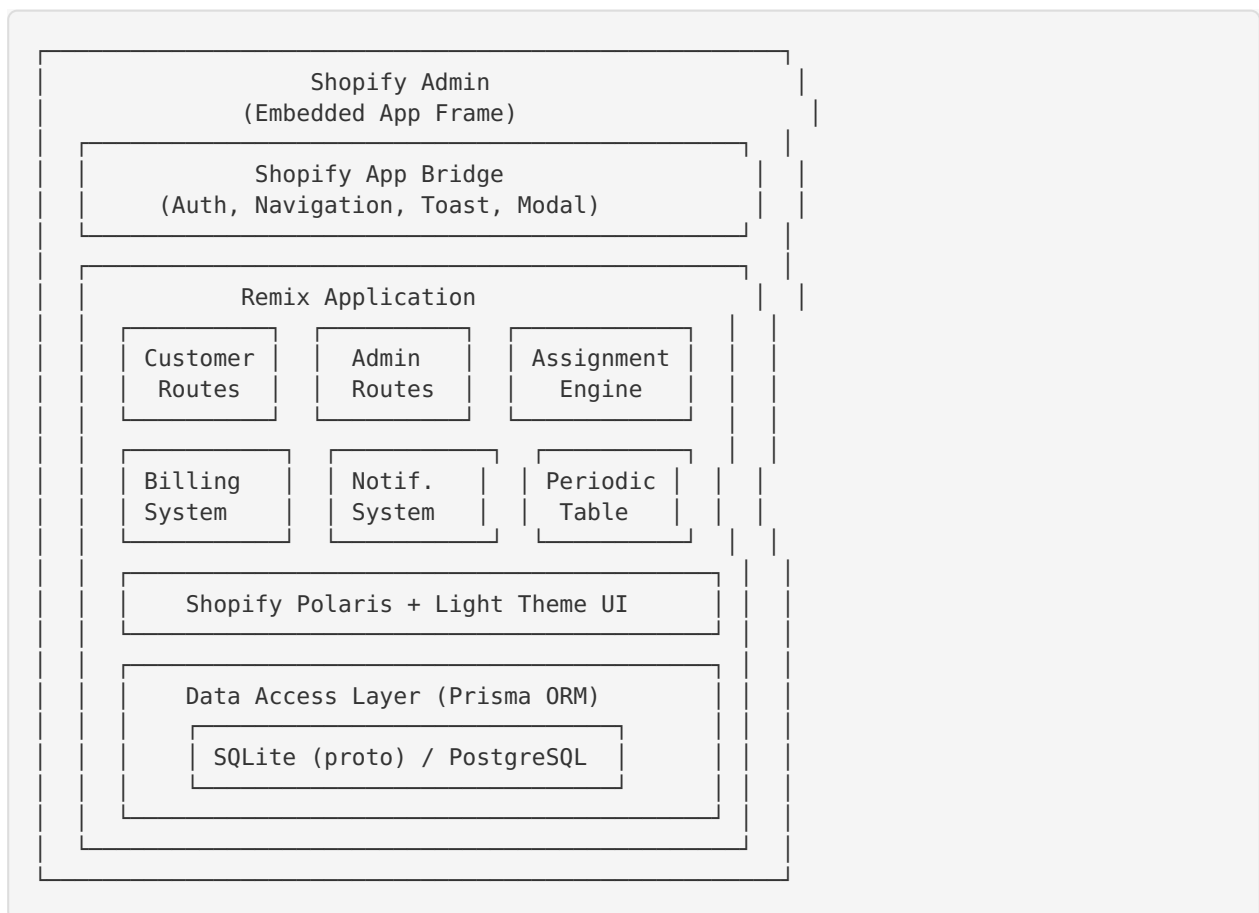
Luciteria Collector Cabinet — Architecture Document

System Overview

The Collector Cabinet is a Shopify embedded app that provides:

1. A customer-facing “digital cabinet” for tracking element collections
2. An interactive periodic table showing all 118 elements with collection status
3. A subscription management system with intelligent, duplicate-free assignments
4. Collection type selection (10mm, 25.4mm, 50mm, Lucite, Ampoules)
5. A billing system with dual billing model and grandfathering
6. An admin operations dashboard with discount monitoring and sequence preview
7. Email/webhook notification stubs for 8 event types

Technology Stack



Sitemap

Customer-Facing Pages

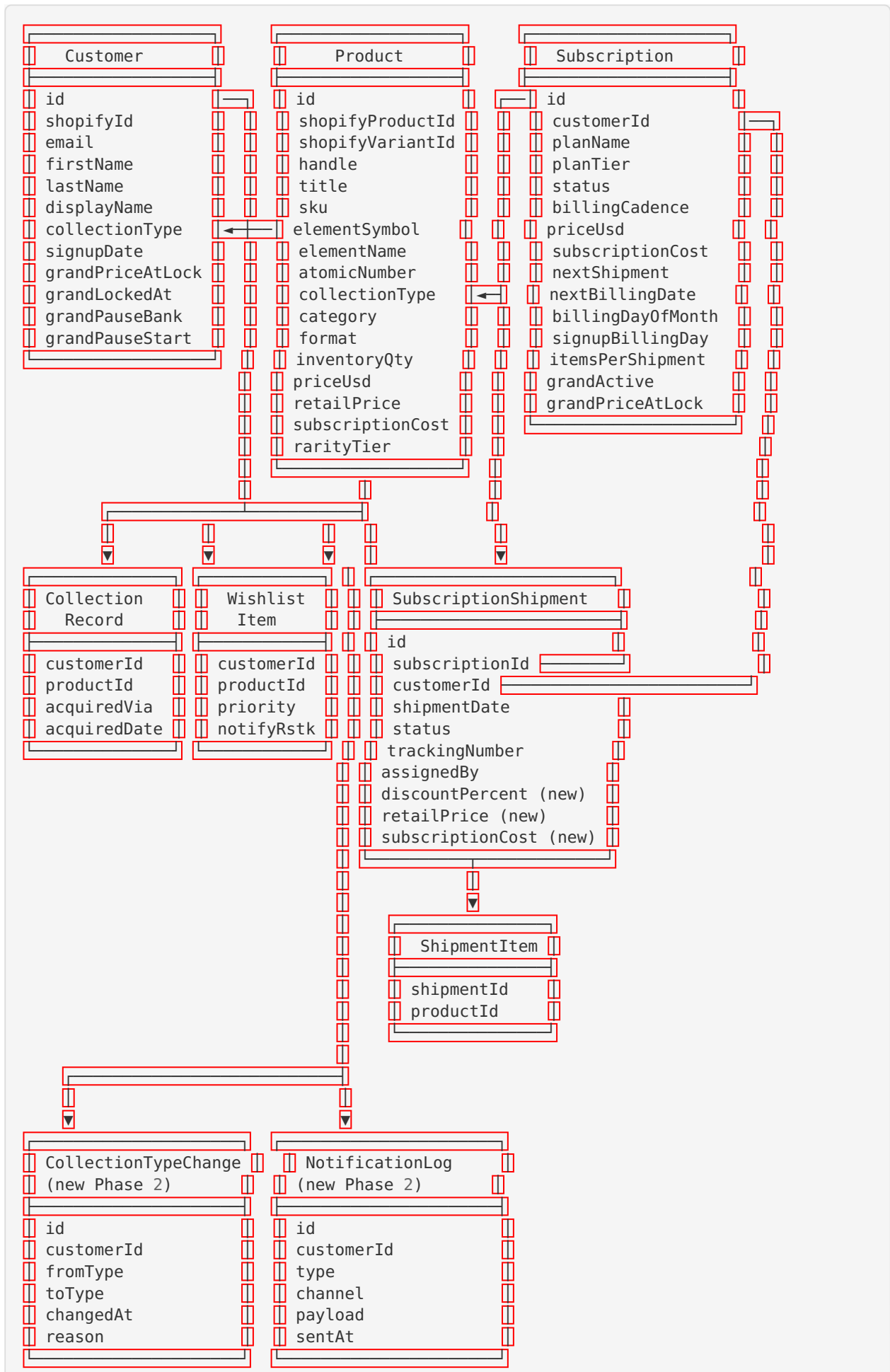
/app/cabinet	→ Dashboard (periodic table, collection summary, recent shipments)
/app/cabinet/collection	→ Collection grid filtered by collection type
/app/cabinet/missing	→ Missing items with collection type awareness
/app/cabinet/wishlist	→ Wishlist management
/app/cabinet/subscription	→ Subscription plan, billing schedule, shipment history
/app/cabinet/preferences	→ Assignment preferences & notification settings
/app/cabinet/onboarding	→ Collection type selection (new subscribers)

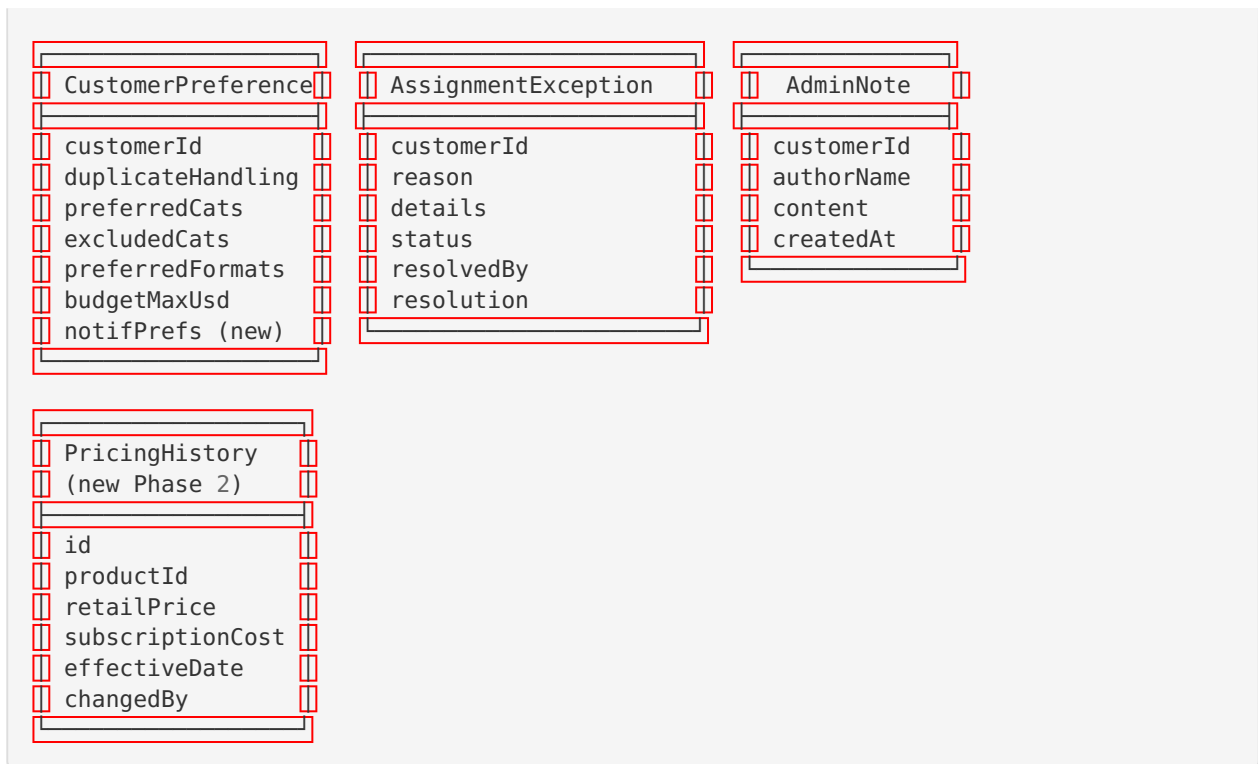
Admin Pages

/app/admin/operations	→ Operations dashboard (discount monitoring, sequence preview)
/app/admin/customers	→ Customer list with collection type badges
/app/admin/customer/:id	→ Individual customer profile with periodic table

Data Model

Entity Relationship Diagram (Phase 2)





New Phase 2 Fields

Entity	New Field	Purpose
Customer	collectionType	Selected collection format (10mm/25.4mm/50mm/lucite/ampoules)
Customer	signupDate , grandPriceAt-Lock , etc.	Grandfathering tracking
Product	collectionType	Which collection format this product belongs to
Product	retailPrice , subscription-Cost	Dual pricing for discount calculations
Subscription	nextBillingDate , billing-DayOfMonth	Billing schedule fields
Subscription	grandActive , grandPriceAt-Lock	Grandfathered pricing
SubscriptionShipment	discountPercent , retail-Price , subscriptionCost	Discount tracking per shipment

New Phase 2 Models

Model	Purpose
CollectionTypeChange	Audit log when customer changes collection type
PricingHistory	Track price changes over time
NotificationLog	Log all notification dispatches

Billing System Architecture

Dual Billing Model

Customer subscribes → Two billing events:

1. Signup Day Billing: Charged on the same day-of-month they signed up
2. 1st-of-Month Billing: Charged on the 1st for the upcoming month's element

5-Day Edge Case:

If signup is within 5 days of month end (e.g., Jan 28):

- First element ships immediately
- Next billing starts Feb 1 (not Feb 28)
- Avoids double-charge within days

Grandfathering Clock

Customer pauses subscription:

- Pause start date recorded
- Grandfathering clock pauses (pause bank accumulates)

Customer resumes:

- Pause bank = days paused
- Grandfathering expiry extended by pause bank days
- Price stays locked at grandPriceAtLock

Notification System Architecture

Event Types

- | | |
|-------------------------|--------------------------------------|
| 1. assignment_preview | ☐ "Here's what's shipping next" |
| 2. shipment_confirmed | ☐ "Your element has shipped" |
| 3. restock_alert | ☐ "Wishlist item back in stock" |
| 4. discount_alert | ☐ "Admin: discount > 20% detected" |
| 5. billing_reminder | ☐ "Billing in 3 days" |
| 6. collection_milestone | ☐ "You've reached 50 elements!" |
| 7. pause_confirmation | ☐ "Subscription paused" |
| 8. welcome | ☐ "Welcome to the Collector Cabinet" |

Implementation (Prototype)

All notifications are stubs that:

- Log to console with full payload

- Store in NotificationLog table (when using Prisma)
- Return success response for UI integration

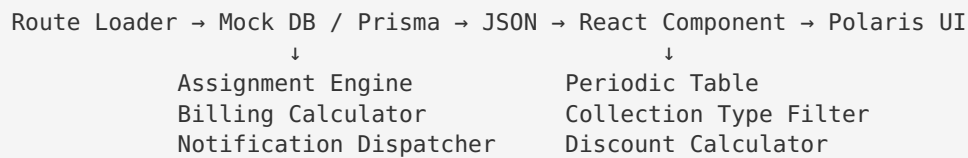
Production would replace stubs with SendGrid/Postmark email and Shopify webhooks.

Component Architecture

Shared Components

- `AppNav` — Sidebar navigation (customer/admin modes) with collection type badge
- `PeriodicTable` — Interactive 118-element periodic table (standard layout, click details, compact mode)
- `StatCard` — Summary metric card (light theme)
- `ProductCard` — Element display card with status badges and discount info
- `ProgressBar` — Collection completion indicator

Data Flow



Theme

- **Phase 1:** Dark theme (cabinet metaphor)
- **Phase 2:** White/light theme (Luciteria branding)
- `--luc-bg: #f5f7fa` (page background)
- `--luc-surface: #ffffff` (card surfaces)
- `--luc-accent: #4A90E2` (primary blue)
- `--luc-text: #1a1a2e` (dark text on light)
- Mobile responsive sidebar (collapsible at 768px)